21-242 Recitation Notes

Connor Gordon

Carnegie Mellon University

Fall 2023

Contents

1 Vector Space Basics **2**
1.1 Logic and Sets . . . 3
1.2 Complex Numbers . . . 4
1.3 Logic and Sets (part 2) . . . 5
1.4 Vector Spaces . . . 6
1.5 Induction . . . 7
1.6 Span . . . 8
1.7 Vector Spaces (part 2) . . . 9
1.8 Linear Independence, Bases, and Span . . . 10
1.9 Fields . . . 11
1.10 Algebraic Field Extensions . . . 12

2 Matrices and Systems of Equations **13**
2.1 Coordinates . . . 14
2.2 Coordinates and Gaussian Elimination . . . 15
2.3 Gaussian Elimination and Kernels . . . 16
2.4 Rank-Nullity . . . 17
2.5 Midterm 1 Review . . . 18

3 Determinants **19**
3.1 Working Towards the Determinant . . . 20
3.2 Determinants and Change of Basis . . . 21

4 Spectral Theory **22**
4.1 Eigenvectors, Eigenvalues, and Diagonalization . . . 23
4.2 Markov Chains . . . 24
4.3 Inner Product Spaces . . . 25
4.4 More Inner Products . . . 26
4.5 Gram-Schmidt and Adjoints . . . 27
4.6 Problems We Didn't Get To . . . 28
4.7 Midterm 2 Review . . . 29

5 Applications of Spectral Theory **30**
5.1 Least Squares and Minimal Norm . . . 31
5.2 Singular Value Decomposition . . . 32
5.3 More Singular Value Decomposition . . . 33
5.4 Jordan Canonical Form . . . 34
5.5 More Jordan Canonical Form . . . 35
5.6 Final Exam Review . . . 36

Chapter 1

Vector Space Basics

1.1 Logic and Sets

1. Given two integers a and b, we say that *a divides b* if there exists some integer k such that $b = ka$.

(a) Show that -3 divides 12.

(b) Show that every integer divides 0, and that 0 does not divide any nonzero integer.

2. Suppose a, b, and c are integers such that a divides b and b divides c. Show that a divides c.

3. We say a positive integer $p > 1$ is *prime* if for any integer $2 \leq k < p$, k does not divide p.

(a) Show that 7 is prime, and that 8 is not prime.

(b) Show that a number is prime if and only if for any *prime* $2 \leq q < p$, q does not divide p. More precisely, you want to show the following two statements are equivalent for any integer $p > 1$:

I. For any integer k such that $2 \leq k < p$, k does not divide p.

II. For any prime q such that $2 \leq q < p$, q does not divide p.

You may use without proof the fact that any integer greater than 2 is divisible by at least one prime number (possibly itself).

4. We say that A is a *subset* of B, denoted $A \subseteq B$, if for all x, if $x \in A$ then $x \in B$. Suppose A is the set of integers divisible by 4 and B is the set of integers divisible by 2. Show that $A \subseteq B$.

5. To show two sets A and B are equal, it suffices to show that $A \subseteq B$ and $B \subseteq A$ (this is known as proof by *double containment*). Let A be the set of integers divisible by 2, let B be the set of integers divisible by 3, and let C be the set of integers divisible by 6. Show that $A \cap B = C$.

(For two sets A and B, the *intersection* of A and B, denoted $A \cap B$, is the set of all elements of both A and B. To show $x \in A \cap B$, it suffices to show $x \in A$ and $x \in B$.)

1.2 Complex Numbers

1. The *complex numbers* are a two-dimensional extension of the real numbers. The first dimension is the *real axis* and the second is the *imaginary axis*. The complex number corresponding to the pair (a, b), where a and b are real numbers, is usually written $a + bi$, where i is the *imaginary unit*. If either a or b is 0, we often just write the nonzero part (unless both are 0, in which case we just write 0).

Complex addition is defined as you would expect:

$$(a + bi) + (c + di) = (a + c) + (b + d)i.$$

It's not hard to see that complex addition is commutative and associative. Find a complex number z such that $(3 + 4i) + z = 0$.

2. Complex multiplication is less straightforward:

$$(a + bi) \cdot (c + di) = (ac - bd) + (ad + bc)i.$$

One can check (don't worry about it; it's tedious) that complex multiplication is commutative, associative, and distributive over addition. A good way to think about complex multiplication is that we can apply the usual distributivity (perhaps you know it as FOIL) rules, where we define $i^2 = -1$. Indeed,

$$(a + bi) \cdot (c + di) = ac + adi + bci + bdi^2 = (ac - bd) + (ad + bc)i.$$

Find a complex number z such that $(1 + i) \cdot z = 2$.

3. Integer complex exponentiation (we'll worry about more general exponentiation later) is the same as for real numbers: repeated multiplication. Compute $i + i^2 + i^3 + \cdots + i^{100}$.

4. **(a)** For a complex number $z = a + bi$, its *complex conjugate* is the number $\bar{z} = a - bi$. Prove that $\overline{z + w} = \bar{z} + \bar{w}$ and $\overline{z \cdot w} = \bar{z} \cdot \bar{w}$.

(b) Prove that for any complex number z, the quantity $z \cdot \bar{z}$ is a non-negative real number.

(c) We define the *magnitude* or *absolute value* of a complex number z, denoted $|z|$, to be $|z| = \sqrt{z \cdot \bar{z}}$. Compute $|a + bi|$ in terms of a and b.

(d) Show that for complex numbers z and w, $|z \cdot w| = |z| \cdot |w|$ (using the definition directly; don't use the previous part).

(e) Show that for any nonzero complex number z, the number $\frac{1}{z} := \frac{\bar{z}}{|z|^2}$ satisfies $z \cdot \frac{1}{z} = 1$.

5. These problems are not particularly important, but they're pretty fun!

(a) Find all complex numbers z such that $z + \frac{1}{z}$ is real.

(b) By considering the product $(a + bi)(c + di)$ and previous results, show the identity

$$(a^2 + b^2)(c^2 + d^2) = (ac - bd)^2 + (ad + bc)^2$$

without expanding everything.

1.3 Logic and Sets (part 2)

1. At the beginning of recitation, I will go over the definitions of the logical operations $\neg$ (not), $\wedge$ (and), $\vee$ (or), $\rightarrow$ (if, then), and $\leftrightarrow$ (if and only if), and how to write truth tables for expressions involving them. Use this information to write truth tables for $\neg p$, $p \wedge q$, $p \vee q$, $p \rightarrow q$, and $p \leftrightarrow q$.

2. We say two logical formulas are *equivalent* if they have the same truth value for all possible variable assignments. Show that the following pairs of formulas are equivalent by drawing truth tables (you don't have to do all of them; pick two or three to do completely and convince yourself the rest also work).

- **(a)** $p \rightarrow q$ and $(\neg p) \vee q$
- **(b)** $p \leftrightarrow q$ and $(p \wedge q) \vee ((\neg p) \wedge (\neg q))$
- **(c)** $\neg(p \vee q)$ and $(\neg p) \wedge (\neg q)$
- **(d)** $\neg(p \wedge q)$ and $(\neg p) \vee (\neg q)$
- **(e)** $p \wedge (q \vee r)$ and $(p \wedge q) \vee (p \wedge r)$
- **(f)** $p \vee (q \wedge r)$ and $(p \vee q) \wedge (p \vee r)$

3. I will also go over the definitions of the set operations $\cap$ (intersection), $\cup$ (union), and $\setminus$ (set subtraction/complement). Prove the following for any sets A and B:

- **(a)** $A \cap B \subseteq A$
- **(b)** $A \subseteq A \cup B$
- **(c)** If $A \cap B = A$, then $A \subseteq B$.
- **(d)** If $A \cup B = A$, then $B \subseteq A$.

4. There is a connection between these logical operations and these set operations! Let's fix a container set S, and for any logical formula p that takes in a parameter x, define $S_p = \{x \in S : p(x) \text{ is true}\}$. We can then see, that, for instance, $S_{p \wedge q} = S_p \cap S_q$. In a similar manner, write expressions for each of the following sets in terms of S, S_p, S_q, and set operations (remember that you can write $S \setminus S_p$ for the complement of S_p).

- **(a)** $S_{\neg p}$
- **(b)** $S_{p \vee q}$
- **(c)** $S_{p \rightarrow q}$
- **(d)** $S_{p \leftrightarrow q}$
- **(e)** S_{true}
- **(f)** S_{false}

5. This suggests a new approach to proving sets are equal! Given two expressions in terms of set operations, we can convert to the corresponding formulas and then compare the truth tables of those formulas. If you've ever seen Venn diagrams before (ask me about it if not), this is essentially what you are doing with those.

Given two sets A and B, we define their *symmetric difference* $A \triangle B$ to be the set of all elements in *exactly* one of A or B. Note that it follows immediately that $A \triangle B = B \triangle A$

- **(a)** Prove, via truth table or Venn diagram, that for any sets A and B

$$A \triangle B = (A \cup B) \setminus (A \cap B) = (A \setminus B) \cup (B \setminus A).$$

- **(b)** Prove that for any sets A, B, and C, $(A \triangle B) \triangle C = A \triangle (B \triangle C)$.
- **(c)** Prove that for any sets A, B, and C, $A \cap (B \triangle C) = (A \cap B) \triangle (A \cap C)$.

1.4 Vector Spaces

1. For each of the following spaces, determine whether or not the space is a vector space over $\mathbb{R}$ when equipped with the natural vector addition and scalar multiplication operations. Justify your answer.

(a) The set of all $n \times n$ matrices (for now, a matrix is just a "box of numbers") with real entries (operations are defined "coordinate-wise").

(b) The set of all *symmetric* $n \times n$ matrices with real entries (a matrix is *symmetric* if it remains the same when swapping the rows and columns).

(c) The set of all anti-symmetric $n \times n$ matrices with real entries (a matrix is *anti-symmetric* if swapping the rows and columns negates the matrix).

(d) The set of all non-negative functions from $[0, 1] \to \mathbb{R}$ (operations are defined pointwise).

(e) For an integer $n \geq 1$, the set of all polynomials of degree *exactly* n with real coefficients.

(f) For an integer $n \geq 1$, the set of all polynomials of degree *at most* n with real coefficients.

2. Consider the space $\mathbb{R} \times \mathbb{C}$, that is, the set of pairs (x, z) with $x \in \mathbb{R}$ and $z \in \mathbb{C}$.

(a) Show that $\mathbb{R} \times \mathbb{C}$ is a vector space over $\mathbb{R}$ with pointwise addition and pointwise scalar multiplication.

(b) Show that $\mathbb{R} \times \mathbb{C}$ is *not* a vector space over $\mathbb{C}$ with pointwise addition and pointwise scalar multiplication.

(c) Consider $\mathbb{R} \times \mathbb{C}$ over $\mathbb{C}$ equipped with pointwise vector addition, but scalar multiplication is defined via

$$\alpha \cdot (x, z) = \begin{cases} (\alpha x, \alpha z) & \alpha \in \mathbb{R} \\ (0, \alpha z) & \alpha \notin \mathbb{R}. \end{cases}$$

Is this a vector space?

3. Suppose $\mathbf{u}$ and $\mathbf{v}$ are vectors such that $\mathbf{u} \neq \alpha \cdot \mathbf{v}$ for all scalars α. Suppose that α, β, γ, and δ are scalars such that $\alpha \cdot \mathbf{u} + \beta \cdot \mathbf{v} = \gamma \cdot \mathbf{u} + \delta \cdot \mathbf{v}$. Show that $\alpha = \gamma$ and $\beta = \delta$.

You may use without proof (only here, not on the homework!) that $\alpha \cdot \mathbf{v} = \mathbf{0}$ if and only if $\alpha = 0$ or $\mathbf{v} = \mathbf{0}$.

4. Let V and W be vector spaces over the same field. A function $T : V \to W$ is called a *linear transformation* if for any vectors $\mathbf{v}$ and $\mathbf{w}$ in V and scalar α,

$$T(\mathbf{v} + \mathbf{w}) = T(\mathbf{v}) + T(\mathbf{w}) \quad \text{and} \quad T(\alpha \cdot \mathbf{v}) = \alpha \cdot T(\mathbf{v}).$$

Suppose T is a linear transformation from V to W.

(a) Show $T(\mathbf{0}_V) = \mathbf{0}_W$.

(b) Suppose $\mathbf{v}$ is a vector in V. Show that $T(-\mathbf{v}) = -T(\mathbf{v})$.
You may take for granted the results of problem 1 and 2 on the homework, namely that additive inverses are unique and that $(-1) \cdot \mathbf{v} = -\mathbf{v}$ (you don't strictly need them, but the proof that doesn't use them is basically equivalent to the proof of them).

5. Let V and W be vector spaces over the same field, and suppose T and S are linear transformations from V to W and α is a scalar.

(a) Show that $T + S$, the transformation mapping $\mathbf{v}$ to $T(\mathbf{v}) + S(\mathbf{v})$, is a linear transformation.

(b) Show that $\alpha \cdot T$, the transformation mapping $\mathbf{v}$ to $\alpha \cdot T(\mathbf{v})$, is a linear transformation.

(c) Conclude that the space of linear transformations from V to W is a vector space over the same field as V and W.

1.5 Induction

1. For each positive integer n, let P_n be the assertion that $1 + 2 + \cdots + n = \frac{n(n+1)}{2}$. We want to prove that P_n is true for every positive integer n. Do so by completing the following steps.

(a) What does P_1 assert? Prove that it is true.

(b) Suppose you know that P_{10} is true; that is, that $1 + 2 + \cdots + 10 = \frac{10(10+1)}{2} = 55$. Now add 11 to both sides. Use this to complete a proof that P_{11} is true.

(c) Now generalize the previous step. For some positive integer k, suppose you know that P_k is true. Use this information to show that P_{k+1} is true.

(d) Use the previous steps to conclude that, for any positive integer n, P_n is true.

This proof technique is called *(weak) induction.* Part (a) is known as the *base case* and part (c) is known as the *induction step*.

2. For each positive integer n, let P_n be the assertion that $1^2 + 2^2 + \cdots + n^2 = \frac{n(n+1)(2n+1)}{6}$. Prove that P_n is true for every positive integer n by induction.

3. Consider the following "proof" that for any group of horses, all of the horses are the same color.

We proceed by induction on the number n of horses in the group. Let P_n be the assertion that in any group of horses of size n, all of the horses are the same color. The base case P_1 is obvious. For the induction step, fix a positive integer k and suppose P_k is true. Consider an arbitrary group of $k + 1$ horses, and label the horses $1, 2, \ldots, k + 1$. By P_k, the group $1, \ldots, k$ are all the same color. Also by P_k, the group $2, \ldots, k + 1$ are all the same color. These two together imply that the combined group of $k + 1$ horses are all the same color. Since we considered an arbitrary group of $k + 1$ horses, this proves P_{k+1}. By induction, we are done.

But obviously not all horses are the same color! Where's the mistake in the proof?

4. For each positive integer $n \geq 2$, let P_n be the assertion that n is divisble by at least one prime number (we call this number a *prime factor* of n).

(a) Prove that P_2 is true.

(b) Suppose you know that all of $P_2, P_3, \ldots, P_{209}$ are true, and have noticed that 210 is divisible by 6, namely $210 = 6 \cdot 35$. Use this observation and the assumptions to conclude that P_{210} is true.

(c) Now we generalize the previous part. For some positive integer k, suppose you know that $P_2, \ldots, P_k$ are all true. Show that if $k + 1$ is prime, then we are done.

(d) If $k + 1$ is not prime, then there is some $2 \leq m < k + 1$ that divides $k + 1$. Show that in this case we are also done using one of the assumed P_i's.

(e) Use the previous steps to conclude that, for any positive integer n, P_n is true.

This proof technique, where we assume all smaller cases in our induction step instead of just one, is called *strong induction*.

5. The following two parts are known as the *Fundamental Theorem of Arithmetic*.

(a) Prove that every positive integer $n \geq 2$ can be factored into a product of primes.

(b) Show that, moreover, this factorization is unique. That is, any factorization of a fixed n consists of the same prime numbers. The following lemma, which you may assume without proof, may be of use.

For any prime p and integers a and b, if p divides ab, then either p divides a or p divides b.

1.6 Span

1. For each of the following vector spaces V over $\mathbb{F}$, explicitly construct a size n list of vectors $\mathcal{L}$ in V such that $\text{Span}(\mathcal{L}) = V$.

(a) V is the set of vectors $\langle x, y, z \rangle \in \mathbb{F}^3$ such that $x + y + z = 0$, $n = 2$.

(b) V is the set of 3×3 symmetric matrices with entries in $\mathbb{F}$, $n = 6$.

(c) V is the set of 3×3 anti-symmetric matrices with entries in $\mathbb{F}$, $n = 3$.

2. Let V be a vector space and $A, B \subseteq V$.

(a) Prove that $A \subseteq \text{Span}(A)$.

(b) Prove that if $A \subseteq B$, $\text{Span}(A) \subseteq \text{Span}(B)$.

These two properties, along with 3(b), show that Span is what's known as a *closure operator*.

3. Let V be a vector space over $\mathbb{F}$ and $A \subseteq V$.

(a) Show that $\text{Span}(A) = A$ if and only if for all $\alpha \in \mathbb{F}$ and $\mathbf{v}, \mathbf{w} \in A$, $\alpha\mathbf{v} + \mathbf{w} \in A$.
(Hint: At some point you'll want to use induction on the length of a linear combination.)

(b) Deduce that $\text{Span}(\text{Span}(A)) = \text{Span}(A)$.

4. Let V be a vector space over $\mathbb{F}$ and $A \subseteq V$. Prove that $\text{Span}(A)$ is the smallest set containing A that is closed under $+$ and $\cdot$. That is, if B is some set such that $A \subseteq B$ and $\forall \alpha \in \mathbb{F}$, $\forall \mathbf{v}, \mathbf{w} \in B$, $\alpha\mathbf{v} + \mathbf{w} \in B$, then $\text{Span}(A) \subseteq B$.

5. Let $P_n(\mathbb{R})$ be the set of polynomials of degree at most n with real coefficients, viewed as a vector space over $\mathbb{R}$. Prove that if p is a polynomial with degree n, then $\{p(x), p(x+1), \ldots, p(x+n)\}$ spans $P_n(\mathbb{R})$.
(Hint: Induct on n. Consider the polynomials $p(x+1) - p(x)$, $p(x+2) - p(x+1)$, and so on.)

1.7 Vector Spaces (part 2)

1. Let V be a vector space over $\mathbb{F}$ and A be a set. We define the set

$$^{A}V = \{f : f \text{ is a function from } A \text{ to } V\}.$$

Show that ^{A}V is a vector space over $\mathbb{F}$ with pointwise operations.

2. Suppose $(V, \oplus_V, \odot_V)$ is a vector space over $\mathbb{F}$, and suppose $f : V \to W$ is a bijection for some set W. Show that if we define the operations

$$\mathbf{u} \oplus_W \mathbf{v} = f(f^{-1}(\mathbf{u}) \oplus_V f^{-1}(\mathbf{v}))$$
$$\alpha \odot_W \mathbf{v} = f(\alpha \odot_V f^{-1}(\mathbf{v})),$$

then $(W, \oplus_W, \odot_W)$ is a vector space over $\mathbb{F}$.

3. Suppose we wanted to consider $\mathbb{R}$ as a vector space over $\mathbb{C}$. Further suppose that we want vector addition and scalar multiplication to behave with the structure of $\mathbb{R}$, namely $x \oplus y = x + y$ and for all real α, $\alpha \odot y = \alpha y$. Is there a way to define scalar multiplication for non-real α such that $(\mathbb{R}, \oplus, \odot)$ is a vector space over $\mathbb{C}$? Justify your answer.

4. If we are not beholden to the arithmetic structure of the reals, then we can consider $\mathbb{R}$ as a vector space over $\mathbb{C}$! Given that there exists a bijection $\mathbb{R} : \mathbb{R} \to \mathbb{C}$,[1] use this bijection and problem 2 to define $\oplus$ and $\odot$ such that $(\mathbb{R}, \oplus, \odot)$ is a vector space over $\mathbb{C}$.

5. For this problem, we will consider only structures $(V, \oplus)$ with only an addition operation.

(a) Find a structure $(V, \oplus)$ such that $\oplus$ is associative but not commutative.

(b) Find a structure $(V, \oplus)$ such that $\oplus$ is commutative and associative, but there is no additive identity.

(c) Find a structure $(V, \oplus)$ such that $\oplus$ is commutative and associative, and there is an additive identity, but there are not additive inverses.

[1] something something first biject $\mathbb{R}$ with $\mathbb{R}^2$ with alternating digits in binary expansion, then biject $\mathbb{R}^2$ with $\mathbb{C}$ in the natural way

1.8 Linear Independence, Bases, and Span

1. Consider the set of all sequences $a_0, a_1, a_2, \ldots$ of real numbers such that for all $n \geq 2$, $a_n = a_{n-1} + a_{n-2}$.

(a) Verify that this is a vector space over $\mathbb{R}$ when equipped with pointwise operations.

(b) Find an explicit basis of this space.

2. Let V be a vector space, and all of the following lists be lists of vectors in V.

(a) Suppose $L = \{\mathbf{v}_1, \ldots, \mathbf{v}_k\}$ is linearly independent. Show that any subset of L is also linearly independent.

(b) Suppose $S = \{\mathbf{v}_1, \ldots, \mathbf{v}_k\}$ is spanning. Show that any superset of S is spanning.

(c) Suppose that $B = \{\mathbf{v}_1, \ldots, \mathbf{v}_k\}$ is a basis. Show that no proper subset or proper superset of B is a basis.

3. Let V be a vector space over $\mathbb{F}$ and $A \subseteq V$. Prove that $\text{Span}(A)$ is the smallest set containing A that is closed under $+$ and $\cdot$. That is, if B is some set such that $A \subseteq B$ and $\forall \alpha \in \mathbb{F}, \forall \mathbf{v}, \mathbf{w} \in B$, $\alpha\mathbf{v} + \mathbf{w} \in B$, then $\text{Span}(A) \subseteq B$.

4. Let V and W be vector spaces over $\mathbb{F}$, and recall that a function $T : V \to W$ is a *linear transformation* if for all $\alpha \in \mathbb{F}, \mathbf{v}, \mathbf{w} \in V$,

$$T(\mathbf{v} + \mathbf{w}) = T(\mathbf{v}) + T(\mathbf{w}) \text{ and } T(\alpha\mathbf{v}) = \alpha T(\mathbf{v}).$$

Let $\{\mathbf{v}_1, \ldots, \mathbf{v}_n\}$ be a basis of V, and suppose $T_1, T_2 : V \to W$ are linear transformations such that $T_1(\mathbf{v}_i) = T_2(\mathbf{v}_i)$ for all $i = 1, \ldots, n$. Show that $T_1(\mathbf{v}) = T_2(\mathbf{v})$ for all $\mathbf{v} \in V$.

5. Let V and W be vector spaces over $\mathbb{F}$, and suppose $T : V \to W$ is a linear transformation. Let $A = \{\mathbf{v}_1, \ldots, \mathbf{v}_k\}$ be a list of vectors in V.

(a) We say T is *injective* if for all $\mathbf{v}, \mathbf{w} \in V$, if $T(\mathbf{v}) = T(\mathbf{w})$ then $\mathbf{v} = \mathbf{w}$. Show that if T is injective and A is linearly independent in V, then the list $\{T(\mathbf{v}_1), \ldots, T(\mathbf{v}_k)\}$ is linearly independent in W.

(b) We say T is *surjective* if for all $\mathbf{w} \in W$, there exists some $\mathbf{v} \in V$ such that $T(\mathbf{v}) = \mathbf{w}$. Show that if T is surjective and A spans V, then the list $\{T(\mathbf{v}_1), \ldots, T(\mathbf{v}_k)\}$ spans W.

(c) We say T is *bijective* if it is both injective and surjective. Show that if T is bijective and A is a basis of V, then the list $\{T(\mathbf{v}_1), \ldots, T(\mathbf{v}_k)\}$ is a basis of W.

1.9 Fields

1. A *field* is a set of "numbers" equipped with addition, subtraction, multiplication, and division operations that behave in the way you'd expect them to. More precisely, for any x, y, z in the field:

- $x + y = y + x$
- $x + (y + z) = (x + y) + z$
- $(x + y)z = xz + yz$
- $x \cdot y = y \cdot x$
- $x \cdot (y \cdot z) = (x \cdot y) \cdot z$
- $x(y + z) = xy + xz$

There also exist an *additive identity* 0, a *multiplicative identity* 1 (with $0 \neq 1$), an *additive inverse* $-x$ for every x, and a *multiplicative inverse* x^{-1} for every nonzero x such that

- $x + 0 = 0 + x = x$
- $x + (-x) = (-x) + x = 0$
- $x \cdot 1 = 1 \cdot x = x$
- $x \cdot (x^{-1}) = (x^{-1}) \cdot x = 1$

(a) Convince yourself that $\mathbb{Q}$ (the rational numbers), $\mathbb{R}$ (the real numbers), and $\mathbb{C}$ (the complex numbers) are all fields with their usual operations.

(b) Show that $\mathbb{Z}$ (the set of integers) is not a field with its usual operations.

2. Given a field K and a $F \subseteq K$, we say that F is a *subfield* of K if the restrictions of $+_K$ and $\cdot_K$ to F are well-defined.

(a) Convince yourself that $\mathbb{Q}$ is a subfield of $\mathbb{R}$, both of which are subfields of $\mathbb{C}$.

(b) Informally argue that, in order to prove that F is a subfield of K, it suffices to check that F is closed under $+_K$ and $\cdot_K$ and that for any $a \in F$, $-a$ and $\frac{1}{a}$ (as computed in K), are also in F.

(c) Suppose L is a field and F, K are both subfields of L. Show that $F \cap K$ is also a subfield of L (and indeed, an arbitrarily large intersection of subfields remains a subfield).

3. **(a)** Suppose that F and K are fields such that F is a subfield of K. Show that K can be viewed as a vector space over F.

(b) Suppose that F, K, and L are fields such that $F \subseteq K \subseteq L$ (and these inclusions are subfield relations). Let $\{\alpha_1, \ldots, \alpha_{d_1}\}$ be a basis of K over F, and let $\{\beta_1, \ldots, \beta_{d_2}\}$ be a basis of L over K. Show that the set $\{\alpha_i \beta_j : 1 \leq i < d_1, 1 \leq j < d_2\}$ is a basis of L over F.

4. As it turns out, we can actually have finite fields!

(a) Show that the integers mod 5 form a field under standard operations (don't worry about checking all of the axioms; the main ones of interest are additive inverses and multiplicative inverses.

(b) Show that, in fact, for any prime p, the integers mod p form a field under standard axioms. For multiplicative inverses, the following fact may be of use.

(Bézout's Lemma). If m and n are integers and $d = \gcd(m, n)$, then there exist integers a and b such that $am + bn = d$.

(c) Show that for any non-prime $n \geq 2$, the integers mod n do *not* form a field under standard operations.

5. A key property of the finite fields in the previous problem, and in fact all finite fields, is that there exists some integer n such that $1 + 1 + \cdots + 1$ (n times) equals 0. For the integers mod p, that value is p. This value is known as the *characteristic* of the field. If no such integer exists (such as for $\mathbb{Q}$, $\mathbb{R}$, or $\mathbb{C}$), then we say the field has characteristic 0.

(a) Show that there exists a field of characteristic n if and only if n is 0 or prime.

(b) Show that if a field $\mathbb{F}$ has characteristic 0, then $\mathbb{Q} \subseteq \mathbb{F}$. (More precisely, there exists some field operation-preserving bijection from $\mathbb{Q}$ onto a subset of $\mathbb{F}$, which we may call an *embedding* of $\mathbb{Q}$ into $\mathbb{F}$). In this sense we say that $\mathbb{Q}$ is the "smallest" field of characteristic 0.

1.10 Algebraic Field Extensions

1. You've almost certainly seen a proof that $\sqrt{2} \notin \mathbb{Q}$, but what if we wanted to construct a field (say, a subfield of $\mathbb{R}$) that contains both $\mathbb{Q}$ and $\sqrt{2}$? Let's investigate...

(a) Consider the set of all numbers of the form $q_1 + q_2\sqrt{2}$, where q_1 and q_2 are rational. Show that this forms a subfield of $\mathbb{R}$.

(b) Show that this field, which we will call $\mathbb{Q}(\sqrt{2})$, can be viewed as a vector space over $\mathbb{Q}$.

(c) Show that the list $\{1, \sqrt{2}\}$ forms a basis of $\mathbb{Q}(\sqrt{2})$, and thus $\mathbb{Q}(\sqrt{2})$ has dimension 2 as a vector space over $\mathbb{Q}$. Contemplate the fact that the polynomial of minimal degree with leading coefficient 1 and $\sqrt{2}$ as a root is $x^2 - 2$, which has degree 2.

2. We say a number $\alpha \in \mathbb{R}$ is *algebraic* if there is a polynomial $p(x)$ with rational coefficients such that $p(\alpha) = 0$.

(a) Show that if α is algebraic, there is a unique polynomial $p(x)$ with leading coefficient 1 and minimal degree witnessing $p(\alpha) = 0$. This is known as the *minimal polynomial* of α.

(b) Show that if α is algebraic, so is $-\alpha$.

(c) Show that if α is algebraic, so is $\frac{1}{\alpha}$.

3. Let α be an algebraic number, and let d be the degree of its minimal polynomial.

(a) Consider the set of all numbers of the form $q_0 + q_1\alpha + q_2\alpha^2 + \cdots + q_{d-1}\alpha^{d-1}$, where all q_i are rational. Convince yourself that this forms a subfield of $\mathbb{R}$ (just trust me for multiplicative inverses, the same technique that worked for $\sqrt{2}$ works in general, but it's really annoying).

(b) Show that this field, which we will call $\mathbb{Q}(\alpha)$, can be viewed as a vector space over $\mathbb{Q}$.

(c) Show that the list $\{1, \alpha, \alpha^2, \ldots, \alpha^{d-1}\}$ forms a basis of $\mathbb{Q}(\alpha)$, and thus $\mathbb{Q}(\alpha)$ has dimension d as a vector space over $\mathbb{Q}$.

4. Let's more precisely capture what's going on here. Given some "base field" F, a larger "container field" $K \supseteq F$, and some $\alpha \in K$, we define the *field extension* $F(\alpha)$ to be the smallest subfield of K such that $F \subseteq F(\alpha)$ and $\alpha \in F(\alpha)$ (where F_1 is "smaller" than F_2 if $F_1 \subseteq F_2$).

(a) Convince yourself that $\mathbb{Q}(\sqrt{2})$ and $\mathbb{Q}(\alpha)$ as defined in problems 1 and 3 fit this definition.

(b) Show that this definition is well-defined; that is, there exists a subfield of K containing both F and α that is contained within any other such subfield.

(c) Show that if $F(\alpha)$ has finite dimension as a vector space over F, then α is algebraic over F.

5. Now let's tie everything together! Suppose $\alpha, \beta \in \mathbb{R}$ are algebraic over $\mathbb{Q}$. We'll write $\mathbb{Q}(\alpha, \beta)$ as a shorthand for $(\mathbb{Q}(\alpha))(\beta)$.

(a) Show that $\mathbb{Q}(\alpha, \beta)$ has finite dimension as a vector space over $\mathbb{Q}$.

(b) Show that $\mathbb{Q}(\alpha + \beta)$ and $\mathbb{Q}(\alpha \cdot \beta)$ can be viewed as subspaces of $\mathbb{Q}(\alpha, \beta)$. Conclude that $\mathbb{Q}(\alpha + \beta)$ and $\mathbb{Q}(\alpha \cdot \beta)$ are finite-dimensional vector spaces over $\mathbb{Q}$ (you may cite a result from your homework without proof).

(c) Conclude that $\alpha + \beta$ and $\alpha \cdot \beta$ are algebraic over $\mathbb{Q}$. Use this and previous problems to conclude that the algebraic numbers form a field!

Chapter 2

Matrices and Systems of Equations

2.1 Coordinates

1. For each of the following linear maps from $\mathbb{R}^2$ to $\mathbb{R}^2$, compute the corresponding matrix (with respect to standard coordinates).

(a) For $\theta \in [0, 2\pi)$, the map R_θ that rotates all points by an angle θ counterclockwise around the origin.

(b) For $m \in [0, \infty)$, map T_m that reflects all points over the line $y = mx$.

2. Consider the linear map $T : \mathbb{R}^3 \to \mathbb{R}^3$ such that $T(\langle v_1, v_2, v_3 \rangle) = \frac{1}{3}\langle v_2 + v_3, v_1 + v_3, v_1 + v_2 \rangle$. Compute

(a) $[T]$ for the standard coordinates on $\mathbb{R}^3$.

(b) $[T]_{\mathcal{L}}^{\mathcal{L}}$ for the list $\mathcal{L} = (\langle 0, 1, 1 \rangle, \langle 1, 0, 1 \rangle, \langle 1, 1, 0 \rangle)$

3. Consider the following list $\mathcal{L}$ of Lagrange polynomials in $P_2(\mathbb{R})$: $\left(\frac{(x-1)(x-2)}{(0-1)(0-2)}, \frac{(x-0)(x-2)}{(1-0)(1-2)}, \frac{(x-0)(x-1)}{(2-0)(2-1)} \right)$.

(a) Compute coordinate representations for $(x-1)(x-2)$, $(x-1)$, and $(x-2)^2$ with respect to $\mathcal{L}$.

(b) Use this to determine if $(x-1)(x-2)$ is in Span $\left((x-1), (x-2)^2\right)$.

4. Consider the list $\mathcal{L} = (x^2, (x+1)^2, (x+2)^2)$ in $P_2(\mathbb{R})$. Compute the coordinates $[ax^2 + bx + c]_{\mathcal{L}}$.

5. In $\mathbb{R}^2$, let $\mathcal{L}_1 = (\langle 1, 0 \rangle, \langle 0, 1 \rangle)$ and $\mathcal{L}_2 = (\langle 1, 1 \rangle, \langle 1, -1 \rangle)$. If $I : \mathbb{R}^2 \to \mathbb{R}^2$ is the identity map, compute the matrices $[I]_{\mathcal{L}_1}^{\mathcal{L}_2}$ and $[I]_{\mathcal{L}_2}^{\mathcal{L}_1}$.

2.2 Coordinates and Gaussian Elimination

1. Consider the basis $\mathcal{L} = \{\langle 1,0,0\rangle, \langle 1,1,0\rangle, \langle 1,1,1\rangle\}$ of $\mathbb{R}^3$.

(a) For each standard basis vector $\mathbf{e}_i$, of $\mathbb{R}^3$, compute $[\mathbf{e}_i]_{\mathcal{L}}$.

(b) Let T be the linear transformation $\langle x,y,z\rangle \mapsto \langle x+y+z, y+z, z\rangle$. If we let $\mathcal{E}$ be the standard basis of $\mathbb{R}^3$, compute $[T]_{\mathcal{E}}^{\mathcal{L}}$ and $[T]_{\mathcal{L}}^{\mathcal{E}}$.

2. Suppose $\mathcal{L}_1$ and $\mathcal{L}_2$ are bases of a vector space V

(a) Show that $[I]_{\mathcal{L}_2}^{\mathcal{L}_1}[I]_{\mathcal{L}_1}^{\mathcal{L}_2} = [I]_{\mathcal{L}_1}^{\mathcal{L}_2}[I]_{\mathcal{L}_2}^{\mathcal{L}_1} = [I]$, where I is the identity transformation and $[I]$ is taken with respect to the standard basis. Deduce that $[I]_{\mathcal{L}}^{\mathcal{L}}$ is independent of $\mathcal{L}$.

(b) Suppose $T : V \to V$ is a linear transformation. Show that

$$[T]_{\mathcal{L}_2}^{\mathcal{L}^2} = [I]_{\mathcal{L}_1}^{\mathcal{L}_2}[T]_{\mathcal{L}_1}^{\mathcal{L}_1}[I]_{\mathcal{L}_2}^{\mathcal{L}_1}.$$

3. Consider the linear transformation $T : \mathbb{R}^2 \to \mathbb{R}^2$ that maps each vector $\mathbf{v}$ to its reflection over the line $y = mx$. Compute $[T]$ (with respect to the standard basis).

4. Solve the following systems of equations by Gaussian elimination.

(a) $\begin{bmatrix} 1 & 2 \\ 3 & 9 \end{bmatrix} \mathbf{x} = \begin{bmatrix} -3 \\ -15 \end{bmatrix}$.

(b) $\begin{bmatrix} 1 & 2 & 1 \\ 2 & 5 & -3 \\ 0 & 3 & -4 \end{bmatrix} \mathbf{x} = \begin{bmatrix} -2 \\ 0 \\ 5 \end{bmatrix}$.

5. **(a)** Suppose A is a 3×3 matrix. For each of the following row operations, find a matrix E such that EA is the result of the row operation applied to A.

- Multiplying the first row by 2.
- Swapping the first and second rows.
- Subtracting the first row from the second row.

(b) Deduce that Gaussian elimination can be implemented via matrix multiplication on the left.

(c) Suppose that if there is a sequence of row operations that turns A into the identity matrix I. Show that if that same sequence of row operations is applied to the identity matrix I, the resulting matrix B satisfies $BA = I$.

2.3 Gaussian Elimination and Kernels

1. Find the unique solution to each the following systems by Gaussian elimination.

(a) $\begin{bmatrix} 1 & 2 \\ 3 & 9 \end{bmatrix} \mathbf{x} = \begin{bmatrix} -3 \\ -15 \end{bmatrix}$.

(b) $\begin{bmatrix} 1 & 2 & 1 \\ 2 & 5 & -3 \\ 0 & 3 & -4 \end{bmatrix} \mathbf{x} = \begin{bmatrix} -2 \\ 0 \\ 5 \end{bmatrix}$.

2. Find all solutions to each of the following systems of equations, or prove there is no solution.

(a) $\begin{bmatrix} 1 & 2 & 2 \\ 2 & 4 & 6 \\ 1 & 2 & 3 \end{bmatrix} \mathbf{x} = \begin{bmatrix} 1 \\ 4 \\ 3 \end{bmatrix}$

(b) $\begin{bmatrix} 1 & 2 & 2 \\ 2 & 4 & 6 \\ 1 & 2 & 3 \end{bmatrix} \mathbf{x} = \begin{bmatrix} 1 \\ 4 \\ 2 \end{bmatrix}$

3. For each of the following lists $\mathcal{L}$ of vectors, determine if $\mathbf{v}$ is in the span of $\mathcal{L}$, and find a linear combination representing it if so.

(a) $\mathcal{L} = \left\{ \begin{bmatrix} 2 \\ -1 \\ 4 \end{bmatrix}, \begin{bmatrix} 1 \\ 0 \\ -2 \end{bmatrix} \right\}, \mathbf{v} = \begin{bmatrix} -1 \\ -1 \\ 10 \end{bmatrix}$

(b) $\mathcal{L} = \left\{ \begin{bmatrix} 2 \\ -1 \\ 4 \end{bmatrix}, \begin{bmatrix} 1 \\ 0 \\ -2 \end{bmatrix} \right\}, \mathbf{v} = \begin{bmatrix} 3 \\ -2 \\ 6 \end{bmatrix}$

4. For each of the following matrices, compute a basis of its kernel.

(a) $\begin{bmatrix} 1 & -1 & 0 \\ 1 & 2 & 1 \\ 2 & 1 & 1 \end{bmatrix}$

(b) $\begin{bmatrix} 1 & 2 & -1 & -3 \\ 2 & 4 & -1 & 5 \end{bmatrix}$

5. **(a)** Suppose that there is a sequence of row operations that turns A into the identity matrix I. Show that if that same sequence of row operations is applied to the identity matrix I, the resulting matrix B satisfies $BA = I$. (It is in fact the two-sided inverse of A, but I don't think we have the tools for that quite yet.)

(b) Use the previous part to determine when the matrix $\begin{bmatrix} a & b \\ c & d \end{bmatrix}$ has an inverse, and compute this inverse when it does.

2.4 Rank-Nullity

1. Suppose A and B are $n \times n$ matrices. Show that

(a) $\text{rank}(AB) \leq \text{rank}(B)$

(b) $\text{rank}(AB) \leq \text{rank}(A)$

2. Suppose A and B are $n \times n$ matrices.

(a) Show that $\dim(\ker(AB)) \leq \dim(\ker(A)) + \dim(\ker(B))$.

(b) Deduce *Sylvester's rank inequality:*

$$\text{rank}(A) + \text{rank}(B) - n \leq \text{rank}(AB) \leq \min\{\text{rank}(A), \text{rank}(B)\}.$$

3. Suppose A and B are $n \times n$ matrices such that $\text{rank}(AB) = \text{rank}(B)$. Show that $\ker(A) \cap \text{im}(B) = \{\mathbf{0}\}$.

4. Use the previous problem to argue that if A is an $n \times n$ matrix, there exists a positive integer k such that $\ker(A^k) \cap \text{im}(A^k) = \{\mathbf{0}\}$.

5. Suppose A is an $n \times n$ matrix such that $\ker(A) \cap \text{im}(A) = \{\mathbf{0}\}$. Show that for any $\mathbf{v} \in \mathbb{F}^n$ there exists a unique $\mathbf{u} \in \ker(A)$ and $\mathbf{w} \in \text{im}(A)$ such that $\mathbf{v} = \mathbf{u} + \mathbf{w}$.

2.5 Midterm 1 Review

1. Show that each of the following matrices is *not* invertible.

(a) $\begin{bmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{bmatrix}$

(b) $\begin{bmatrix} 1 & 2 & 3 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix}$

(c) $\begin{bmatrix} 1 & 4 & 1 & 0 \\ 2 & 5 & 0 & 1 \\ 3 & 6 & 1 & 4 \end{bmatrix}$

(d) $\begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \\ 1 & 0 & 1 \end{bmatrix}$

(e) $\begin{bmatrix} 1 & 1 & 0 & 7 \\ 2 & 2 & -1 & 8 \\ 3 & -1 & 2 & 9 \\ 4 & 0 & 1 & 10 \end{bmatrix}$

(f) $\begin{bmatrix} 1 & 0 & 0 & 1 \\ 0 & 1 & 1 & 0 \\ 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 1 \end{bmatrix}$

2. Suppose that $f : V \to W$ is linear.

(a) Show that f is injective if and only if f has a linear left inverse.

(b) Show that f is surjective if and only if f has a linear right inverse.

Note that these two problems together give the homework result that f is an isomorphism if and only if it is a linear bijection.

3. Show that if $T : V \to V$ and $T^2 = T$, then there is a basis $\mathcal{L}$ for V such that $[T]_{\mathcal{L}}^{\mathcal{L}}$ has main diagonal of the form $1, 1, \ldots, 1, 0, 0, \ldots, 0$ and is 0 everywhere else.

4. Show that there does not exist a $B \in \mathbb{R}^{2\times 2}$ such that $B^2 = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$.

(Hint: AFSOC such a B exists; either the columns of B form a basis of $\mathbb{R}^2$ or they don't. Arrive at a contradiction in either case.)

5. Suppose $A \in \mathbb{C}^{n\times n}$ is of the form $\begin{bmatrix} \lambda & 1 & 0 & \ldots & 0 \\ 0 & \lambda & 1 & \ldots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & 0 & \lambda & 1 \\ 0 & 0 & 0 & 0 & \lambda \end{bmatrix}$ for $\lambda \neq 0$ (this is sometimes known as a *Jordan block*, for reasons you'll see later). Show there exists $B \in \mathbb{C}^{n\times n}$ such that $B^2 = A$.

Chapter 3

Determinants

3.1 Working Towards the Determinant

1. Compute the determinants of the following matrices using Gaussian elimination.

(a) $\begin{bmatrix} 1 & 0 & 1 & 0 \\ 0 & 1 & 1 & 0 \\ 1 & 0 & 0 & 1 \\ 0 & 1 & 0 & 1 \end{bmatrix}$ **(b)** $\begin{bmatrix} 1 & 2 & 3 & 4 \\ 1 & 1 & 1 & 1 \\ 0 & 1 & 1 & 0 \\ 0 & 0 & 1 & 1 \end{bmatrix}$ **(c)** $\begin{bmatrix} 6 & 7 & 8 & 1 \\ 2 & 1 & 8 & 3 \\ 1 & 4 & 3 & 2 \\ 1 & 1 & 0 & 1 \end{bmatrix}$

2. We say a matrix A is *upper triangular* if $A_{ij} = 0$ for all $i > j$. Show that the determinant of an upper triangular matrix is equal to the product of its diagonal elements.

3. We say that $D : \mathbb{F}^{n\times n} \to \mathbb{F}$ is *alternating* if

$$D\begin{bmatrix} - & \mathbf{v}_1 & - \\ & \vdots & \\ - & \mathbf{v}_i & - \\ & \vdots & \\ - & \mathbf{v}_j & - \\ & \vdots & \\ - & \mathbf{v}_n & - \end{bmatrix} = -D\begin{bmatrix} - & \mathbf{v}_1 & - \\ & \vdots & \\ - & \mathbf{v}_j & - \\ & \vdots & \\ - & \mathbf{v}_i & - \\ & \vdots & \\ - & \mathbf{v}_n & - \end{bmatrix}$$

and *multilinear* if

$$D\begin{bmatrix} - & \mathbf{v}_1 & - \\ & \vdots & \\ - & \mathbf{v}_i + \lambda\mathbf{w} & - \\ & \vdots & \\ - & \mathbf{v}_n & - \end{bmatrix} = D\begin{bmatrix} - & \mathbf{v}_1 & - \\ & \vdots & \\ - & \mathbf{v}_i & - \\ & \vdots & \\ - & \mathbf{v}_n & - \end{bmatrix} + \lambda D\begin{bmatrix} - & \mathbf{v}_1 & - \\ & \vdots & \\ - & \mathbf{w} & - \\ & \vdots & \\ - & \mathbf{v}_n & - \end{bmatrix}.$$

Show that if D is alternating and multilinear, then D is Gaussian computable.

4. **(a)** Suppose $B \in \mathbb{F}^{n\times m}$, and let I_m denote the $m \times m$ identity matrix. Show that the map

$$A \mapsto \det\begin{bmatrix} A & B \\ 0 & I_m \end{bmatrix}$$

(where $A \in \mathbb{F}^{n\times n}$) is Gaussian computable.

(b) Now fix $A \in \mathbb{F}^{n\times n}$, $B \in \mathbb{F}^{n\times m}$. Show that the map

$$C \mapsto \det\begin{bmatrix} A & B \\ 0 & C \end{bmatrix}$$

(where $C \in \mathbb{F}^{m\times m}$) is Gaussian computable.

5. Use the previous problem to show that for $A \in \mathbb{F}^{n\times n}$, $B \in \mathbb{F}^{n\times m}$, and $C \in \mathbb{F}^{m\times m}$,

$$\det\begin{bmatrix} A & B \\ 0 & C \end{bmatrix} = \det(A)\det(C).$$

3.2 Determinants and Change of Basis

1. If A is an $n \times n$ matrix, we define its *trace* to be the sum of its diagonal elements. In symbols, $\mathrm{tr}(A) = \sum_{i=1}^{n} A_{ii}$. Convince yourself that $\mathrm{tr} : \mathbb{F}^{n\times n} \to \mathbb{F}$ is a linear map. In this problem, we will show that the trace is independent of coordinates, just like the determinant.

(a) Let $B \in \mathbb{R}^{n\times n}$ be fixed. Show that the maps $A \mapsto \mathrm{tr}(AB)$ and $A \mapsto \mathrm{tr}(BA)$ are linear.

(b) Let E_{ij} be the $n \times n$ matrix with 1 in index (i, j) and 0 otherwise, and let $B \in \mathbb{R}^{n\times n}$. Show that $\mathrm{tr}(E_{ij}B) = \mathrm{tr}(BE_{ij})$.

(c) Use the previous two parts to conclude that for all $A, B \in \mathbb{R}^{n\times n}$, $\mathrm{tr}(AB) = \mathrm{tr}(BA)$.[1]

(d) Suppose $\mathcal{L}$ and $\mathcal{M}$ are both bases of $\mathbb{F}^n$. Show that $\mathrm{tr}([A]_{\mathcal{L}}^{\mathcal{L}}) = \mathrm{tr}([A]_{\mathcal{M}}^{\mathcal{M}})$.

2. Let $\mathcal{L} = \{\mathbf{v}_1, \ldots, \mathbf{v}_n\}$ be a basis of $\mathbb{F}^n$. Let $T : \mathbb{F}^n \to \mathbb{F}^n$ be the unique linear transformation that sends $\mathbf{v}_i \to i\mathbf{v}_i$ for all $1 \le i \le n$. Compute $\det([T])$.

3. Let $\mathbf{v} = \langle x, y, z\rangle \in \mathbb{R}^3$ be nonzero. Consider the linear transformation $R_\theta : \mathbb{R}^3 \to \mathbb{R}^3$ that corresponds to rotating each point in $\mathbb{R}^3$ counter-clockwise by an angle θ about $\mathbf{v}$ (where counter-clockwise is determined, say, by the right-hand rule applied to $\mathbf{v}$). Compute $\det([R_\theta])$ and $\mathrm{tr}([R_\theta])$.[2]

4. Let P be a plane in $\mathbb{R}^3$ (i.e. a two-dimensional subspace). Consider the linear transformation $F : \mathbb{R}^3 \to \mathbb{R}^3$ the corresponds to reflecting each point in $\mathbb{R}^3$ about P. Compute $\det([F])$ and $\mathrm{tr}([F])$.

5. Given points $x_0, \ldots, x_n \in \mathbb{F}$, define the *pointwise evaluation map* F to take in some $p \in P_n(\mathbb{F})$ and output $\langle p(x_0), \ldots, p(x_n)\rangle$. We call the matrix representation of F with respect to the standard basis on $P_n(\mathbb{F})$ the *Vandermonde matrix* of $x_0, \ldots, x_n$,

$$V(x_0, \ldots, x_n) = \begin{bmatrix} 1 & x_0 & x_0^2 & \ldots & x_0^n \\ 1 & x_1 & x_1^2 & \ldots & x_1^n \\ 1 & x_2 & x_2^2 & \ldots & x_2^n \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 1 & x_n & x_n^2 & \ldots & x_n^n \end{bmatrix}.$$

Using change of basis with the basis $\mathcal{L} = \{1, x - x_0, (x - x_0)(x - x_1), \ldots, (x - x_0)\ldots(x - x_n)\}$, compute $\det(V(x_0, \ldots, x_n))$.

[1] Note that this is only true for products of two matrices; this fails for larger products.

[2] You don't have to be totally formal for this and the following problem, and may refer more vaguely to geometric notions like "perpendicular" if that is useful.

Chapter 4

Spectral Theory

4.1 Eigenvectors, Eigenvalues, and Diagonalization

1. Diagonalize the following matrices.

(a) $\begin{bmatrix} 1 & 2 \\ 2 & 1 \end{bmatrix}$

(b) $\begin{bmatrix} 1 & 1 & 0 \\ 1 & 2 & 1 \\ 0 & 1 & 1 \end{bmatrix}$

2. Suppose that A is a square matrix such that the sum of the entries in each row of A is equal to k for some $k \in \mathbb{F}$. Show that k is an eigenvalue of A.

3. In this problem, we'll establish some basic properties of eigenvalues.

(a) Let A be a square matrix. Show that the characteristic polynomial of A is equal to the characteristic polynomial of A^T. Deduce that A and A^T have the same eigenvalues. (Note they do *not*, in general, have the same eigenvectors.)

(b) Let A be an upper triangular square matrix. Show that the eigenvalues of A are the diagonal elements of A.

4. In this problem, we will connect eigenvalues to the trace and determinant. Let A be a square matrix.

(a) Show that the product of the eigenvalues of A is equal to $\det(A)$.

(b) Show that the sum of the eigenvalues of A is equal to $\operatorname{tr}(A)$.

5. Let A be a square matrix. Show that $A^2 = A$ if and only if A is diagonalizable and every eigenvalue of A is 0 or 1.

4.2 Markov Chains

1. We say a vector $\mathbf{v} \in \mathbb{R}^n$ is *stochastic* if all of its entries are non-negative and the sum of its entries is 1. We say a matrix $M \in \mathbb{R}^{n \times n}$ is *stochastic* if all of its columns are stochastic vectors.[1]

(a) Suppose $M \in \mathbb{R}^{n \times n}$ and $\mathbf{v} \in \mathbb{R}^n$ are stochastic. Show that $M\mathbf{v}$ is also stochastic.

(b) Suppose $M, N \in \mathbb{R}^{n \times n}$ are stochastic. Show that MN is also stochastic.

2. Suppose M is a stochastic matrix.

(a) Show that 1 is an eigenvalue of M.

(b) Show that all eigenvalues of M have absolute value less than or equal to 1.

(c) Suppose $\mathbf{v}$ is an eigenvector of M with eigenvalue not equal to 1. Show that the sum of coordinates of $\mathbf{v}$ is equal to 0 (and hence $\mathbf{v}$ is not stochastic).

(d) (Hard!) Show that M has a stochastic eigenvector with eigenvalue 1. This is known as the *steady state* vector of M.

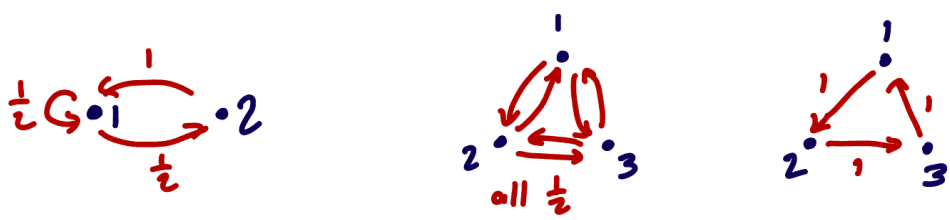
3. A *Markov chain* is a set of states Ω (which we will assume to be finite, but this does not have to be in general), and a set of *transition probabilities* between the states. In particular, we'll let $p(i, j)$ denote the probability that, starting from state i, after one step we end up at state j.

(a) Given a Markov chain, we define its *transition matrix* M to satisfy $M_{ij} = p(j, i)$.[2] Show that M is a stochastic matrix.

(b) Show that $(M^2)_{ij}$ is equal to the probability that, starting at state j, after two steps we end up at state i.

(c) Show (by induction) that for all $n \geq 1$, $(M^n)_{ij}$ is equal to the probability that, starting at state j, after n steps we end up at state i.

4. **(a)** For each of the Markov chains depicted below, compute M and $\lim_{n\to\infty} M^n$, if it exists.



(b) Suppose we know M is diagonalizable. For which values of the eigenvalues of M does $\lim_{n\to\infty} M^n$ converge?

5. Here are some more open-ended questions about Markov Chains.

(a) What does the quantity $(\lim_{n\to\infty} M^n)_{ij}$ represent?

(b) What does the quantity $\text{rank}(\lim_{n\to\infty} M^n)$ represent? If this rank is 1, what does that tell us about the Markov chain?

(c) What does an eigenvalue of -1 mean? Are there other eigenvalues with similar interpretations?

[1] This is the transpose of how things are usually defined. For historical reasons, people generally consider row stochastic vectors and right multiply by stochastic matrices, but given that this class generally does left matrix multiplication on column vectors, I've adjusted these definitions accordingly. If you take a class on Markov chains, you will likely see the other formulation.

[2] Here's that transpose business showing up again... perhaps you can see why people prefer the other way.

4.3 Inner Product Spaces

1. For each of the following matrices $A \in \mathbb{R}^{2\times 2}$, is the map $\langle \mathbf{x}, \mathbf{y} \rangle = \mathbf{y}^T A\mathbf{x}$ an inner product on $\mathbb{R}^2$? Prove your answer.

(a) $A = \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}$ **(b)** $A = \begin{bmatrix} 1 & 2 \\ 2 & 1 \end{bmatrix}$ **(c)** $A = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}$

2. Let V be a finite-dimensional inner product space.

(a) Prove the *Pythagorean theorem* holds in V; that is, if $\mathbf{x}, \mathbf{y} \in V$ such that $x \perp y$, then

$$\|\mathbf{x} + \mathbf{y}\|^2 = \|\mathbf{x} - \mathbf{y}\|^2 = \|\mathbf{x}\|^2 + \|\mathbf{y}\|^2$$

(b) Prove the *parallelogram law* holds in V; that is, if $\mathbf{x}, \mathbf{y} \in V$, then

$$\|\mathbf{x} + \mathbf{y}\|^2 + \|\mathbf{x} - \mathbf{y}\|^2 = 2(\|\mathbf{x}\|^2 + \|\mathbf{y}\|^2)$$

3. Let A be some set, and for $f, g \in \mathbb{F}^A$ consider the map $\langle f, g \rangle = \sum_{a \in A} f(a)\overline{g(a)}$.

(a) Show $\langle \cdot, \cdot \rangle$ is an inner product on $\mathbb{F}^A$.

(b) For $X \subseteq A$, define $\chi_A(x) = \begin{cases} 1 & x \in X \\ 0 & x \notin X \end{cases}$. If $X, Y \subseteq A$, compute $\langle \chi_X, \chi_Y \rangle$.

4. Suppose $A \in \mathbb{R}^{n\times n}$, and that the map $\langle \mathbf{x}, \mathbf{y} \rangle = \mathbf{y}^T A\mathbf{x}$ is an inner product on $\mathbb{R}^n$.

(a) Show that $A^T = A$.

(b) Show that the eigenvalues of A are all real and positive.

5. Suppose a matrix A satisfies $A^T = A$ and has all real and positive eigenvalues. Is the map $\langle \mathbf{x}, \mathbf{y} \rangle$ always an inner product on $\mathbb{R}^n$?

4.4 More Inner Products

1. If $A \in \mathbb{F}^{n\times n}$, its *conjugate transpose* A^* is the matrix obtained by taking the conjugate of every element in A^T.

Let $\mathcal{L} = \mathbf{v}_1, \ldots, \mathbf{v}_k$ be a list of vectors in $\mathbb{F}^n$, and let A be the $n \times k$ matrix such that the ith column of A is $\mathbf{v}_i$. Show that the list $\mathcal{L}$ is orthonormal with respect to the standard inner product on $\mathbb{F}^n$ if and only if A^*A is the $k \times k$ identity matrix.

2. Let V be an inner product space and $\mathbf{x}, \mathbf{y} \in V$. The Cauchy-Schwarz inequality states that $|\langle \mathbf{x}, \mathbf{y}\rangle| \leq \|\mathbf{x}\|\|\mathbf{y}\|$. Show that $|\langle \mathbf{x}, \mathbf{y}\rangle| = \|\mathbf{x}\|\|\mathbf{y}\|$ if and only if one of $\mathbf{x}$ or $\mathbf{y}$ is a scalar multiple of the other. You may use without proof the result from homework that $\|c\mathbf{v}\| = |c|\|\mathbf{v}\|$ for all $c \in \mathbb{F}$ and $\mathbf{v} \in V$.

(Hint: If $\mathbf{y} = 0$ we're done (why?). Otherwise, let $t = \frac{\langle \mathbf{x},\mathbf{y}\rangle}{\|\mathbf{y}\|^2}$, and consider the quantity $\|\mathbf{x} - t\mathbf{y}\|^2$.)

3. In this problem, we will prove what are known as *polarization identities*, which can be used to recover an inner product from a norm.

(a) Let V be a real inner product space and $\mathbf{x}, \mathbf{y} \in V$. Show that

$$\langle \mathbf{x}, \mathbf{y}\rangle = \tfrac{1}{4}(\|\mathbf{x}+\mathbf{y}\|^2 - \|\mathbf{x}-\mathbf{y}\|^2).$$

(b) Let V be a complex inner product space and $\mathbf{x}, \mathbf{y} \in V$. Show that

$$\langle \mathbf{x}, \mathbf{y}\rangle = \tfrac{1}{4}(\|\mathbf{x}+\mathbf{y}\|^2 + i\|\mathbf{x}+i\mathbf{y}\|^2 - \|\mathbf{x}-\mathbf{y}\|^2 - i\|\mathbf{x}-i\mathbf{y}\|^2).$$

4. Let V be a real inner product space. We say $T : V \to V$ *preserves distances* if $\|T(\mathbf{v})\| = \|\mathbf{v}\|$ for all $\mathbf{v} \in V$. We say T *preserves inner products* if $\langle T(\mathbf{v}), T(\mathbf{w})\rangle = \langle \mathbf{v}, \mathbf{w}\rangle$ for all $\mathbf{v}, \mathbf{w} \in V$.

Show that T preserves distances if and only if T preserves inner products.

5. Let V be an inner product space, and let $A \subseteq V$. The *orthogonal complement* of A, denoted $A^\perp$, is defined via $\{\mathbf{v} \in V : \forall \mathbf{a} \in A,\ \mathbf{a} \perp \mathbf{v}\}$.

Show that for any $A \subseteq V$, $A^\perp$ is a subspace of V.

4.5 Gram-Schmidt and Adjoints

1. Use the Gram-Schmidt algorithm on the basis $\left\{ \begin{bmatrix} 2 \\ -2 \\ 0 \end{bmatrix}, \begin{bmatrix} 4 \\ 0 \\ 1 \end{bmatrix}, \begin{bmatrix} 2 \\ 4 \\ 6 \end{bmatrix} \right\}$ of $\mathbb{R}^3$.

2. We define the *Hilbert-Schmidt inner product* on $\mathbb{C}^{n\times n}$ via $\langle A, B\rangle = \operatorname{tr}(B^*A)$.

(a) Show that this is indeed an inner product on $\mathbb{C}^{n\times n}$. You may assume without proof that for any $A \in C^{n\times n}$, A^*A is diagonalizable (this is a useful special case of the spectral theorem).

(b) Compute an orthonormal basis of $\mathbb{C}^{2\times 2}$ with respect to the Hilbert-Schmidt inner product.

3. Let V be an inner product space. We say $T : V \to V$ is *normal* if $TT^* = T^*T$. Suppose T is normal. Show that $\ker(T) = \ker(T^*)$.

4. Suppose T is normal. Show that if $\mathbf{v}$ is an eigenvector of T, it is also an eigenvector of T^* (Hint: First show $T - \lambda I$ is normal for any $\lambda \in \mathbb{F}$).

5. Suppose $A, B \in \mathbb{F}^{n\times n}$. Show that $\ker(A^*A + B^*B) = \ker(A) \cap \ker(B)$.

4.6 Problems We Didn't Get To

1. Determine if each of the following matrices are diagonalizable. Briefly justify your answer. You do not actually need to diagonalize the matrix if it is diagonalizable.

(a) $\begin{bmatrix} 2 & 1 & 1 & 1 \\ 1 & 2 & 1 & 1 \\ 1 & 1 & 2 & 1 \\ 1 & 1 & 1 & 2 \end{bmatrix}$ **(b)** $\begin{bmatrix} 1 & 2 & 3 & 4 \\ 0 & 5 & 6 & 7 \\ 0 & 0 & 8 & 9 \\ 0 & 0 & 0 & 10 \end{bmatrix}$ **(c)** $\begin{bmatrix} 1 & 1 & 1 & 1 \\ 0 & 1 & 1 & 1 \\ 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 1 \end{bmatrix}$ **(d)** $\begin{bmatrix} 1 & 1 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 2 & 1 \\ 0 & 0 & 0 & 2 \end{bmatrix}$

2. Suppose $A \in \mathbb{R}^{n\times n}$, and that the map $\langle \mathbf{x}, \mathbf{y} \rangle = \mathbf{y}^T A\mathbf{x}$ is an inner product on $\mathbb{R}^n$.

(a) Show that $A^T = A$.

(b) Show that the eigenvalues of A are all real and positive.

3. Suppose a matrix A satisfies $A^T = A$ and has all real and positive eigenvalues. Is the map $\langle \mathbf{x}, \mathbf{y} \rangle$ always an inner product on $\mathbb{R}^n$?

4. We define the *Hilbert-Schmidt inner product* on $\mathbb{C}^{n\times n}$ via $\langle A, B \rangle = \operatorname{tr}(B^* A)$.

(a) Show that this is indeed an inner product on $\mathbb{C}^{n\times n}$. You may assume without proof that for any $A \in C^{n\times n}$, A^*A is diagonalizable (this is a useful special case of the spectral theorem).

(b) Compute an orthonormal basis of $\mathbb{C}^{2\times 2}$ with respect to the Hilbert-Schmidt inner product.

5. Suppose $A, B \in \mathbb{F}^{n\times n}$. Show that $\ker(A^*A + B^*B) = \ker(A) \cap \ker(B)$.

4.7 Midterm 2 Review

1. Each part describes a set of matrices. Identify whether the matrices in this set are "always," "sometimes," or "never" diagonalizable.

(a) $A \in \mathbb{R}^{n\times n}$ such that $A^T = A$.

(b) $A \in \mathbb{C}^{n\times n}$ such that $A^T = A$.

(c) $A \in \mathbb{C}^{n\times n}$ such that $AA^* = A^*A$.

(d) $A \in \mathbb{C}^{n\times n}$ such that A has all distinct eigenvalues.

(e) $A \in \mathbb{C}^{n\times n}$ such that A is upper triangular but not diagonal, and all diagonal entries are equal.

2. Let W be the subspace of $\mathbb{R}^4$ consisting of vectors that have a sum of coordinates equal to 0. Find an orthonormal basis of W.

3. Let V be an inner product space, and suppose $\{\mathbf{w}_1, \ldots, \mathbf{w}_n\}$ is an orthonormal basis of V.

(a) Show that for all $\mathbf{v} \in V$, $\mathbf{v} = \sum_{i=1}^n \langle \mathbf{v}, \mathbf{w}_i \rangle \mathbf{w}_i$.

(b) Deduce that for all $\mathbf{v} \in V$, $\|\mathbf{v}\|^2 = \sum_{i=1}^n |\langle \mathbf{v}, \mathbf{w}_i \rangle|^2$.

(c) Suppose V is a subspace of some larger inner product space U of dimension $m \geq n$. Show that one can find vectors $\mathbf{w}_{n+1}, \ldots, \mathbf{w}_m \in U$ such that the list $\{\mathbf{w}_1, \ldots, \mathbf{w}_m\}$ is an orthonormal basis of U.

4. Let V be an inner product space, and let $W \subseteq V$ be a subspace. Let P and Q be the transformations mapping $\mathbf{v} \in V$ to $\text{proj}_W^\perp(\mathbf{v})$ and $\text{proj}_{W^\perp}^\perp(\mathbf{v})$, respectively.

(a) What are $P + Q$, PQ, and QP?

(b) Show that $P - Q$ is its own inverse.

5. Suppose A is a stochastic matrix (recall this means each column of A is a stochastic vector, which is a vector of non-negative real entries which sum to 1).

(a) Show that all entries of A^TA are between 0 and 1 (inclusive).
(Hint: first show that a stochastic vector has norm less than or equal to 1.)

(b) If A is $n \times n$, show that all eigenvalues of A^TA are real and between 0 and n.
(Hint: if $\mathbf{v}$ is an eigenvector, let $\mathbf{v}_j$ be its coordinate with maximal absolute value. Compare $(A^TA\mathbf{v})_j$ to $\mathbf{v}_j$.)

Chapter 5

Applications of Spectral Theory

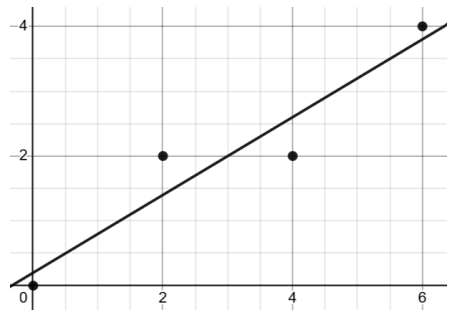
5.1 Least Squares and Minimal Norm

1. Find the least-squares solutions to the following equations.

(a) $\begin{bmatrix} 1 & 1 \\ 1 & -1 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 3 \\ 3 \\ 6 \end{bmatrix}$

(b) $\begin{bmatrix} 1 & i \\ 0 & -i \\ 3i & 1 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 2 \\ -1 \\ 2i \end{bmatrix}$

2. Find the least-squares line fitting the points $(0,0)$, $(2,2)$, $(4,2)$, and $(6,4)$.



3. Suppose $A \in \mathbb{C}^{m \times n}$ is such that $\ker(A) = \{\mathbf{0}\}$. Show that for any $\mathbf{b} \in \mathbb{C}^n$,

$$\text{proj}^{\perp}_{\text{im}(A)}(\mathbf{b}) = A(A^*A)^{-1}A^*\mathbf{b},$$

and deduce that the matrix representation of projection onto $\text{im}(A)$ is given by $A(A^*A)^{-1}A^*$.

(Hint: Recall that $A\mathbf{x} = \text{proj}^{\perp}_{\text{im}(A)}(\mathbf{b})$ if and only if $\mathbf{x}$ is a least-squares solution to $A\mathbf{x} = \mathbf{b}$.)

4. Show that the pseudoinverse of a diagonal matrix is given by inverting all nonzero diagonal entries and leaving zero entries as-is.

5. For $A \in \mathbb{C}^{n \times n}$, we define the *2-norm* of A to be $\|A\|_2 = \sqrt{\sum_{i,j}(A_{ij})^2}$, and we define the *operator norm* of A, denoted $\|A\|_{\text{op}}$, to be greatest value of $\|A\mathbf{v}\|$ over all $\mathbf{v} \in \mathbb{C}^n$ such that $\|\mathbf{v}\| = 1$ (you may take for granted the fact that this quantity is well-defined).

(a) Show that for any $A \in \mathbb{C}^{n \times n}$ and any $r > 0$, there is some invertible $B \in \mathbb{C}^{n \times n}$ such that $\|A - B\|_2 < r$.

(b) Show that for any $A \in \mathbb{C}^{n \times n}$ and any $r > 0$, there is some invertible $B \in \mathbb{C}^{n \times n}$ such that $\|A - B\|_{\text{op}} < r$.

5.2 Singular Value Decomposition

1. Consider the matrices

$$A = \begin{bmatrix} 5 & 1 \\ -1 & -5 \\ -1 & -5 \\ 5 & 1 \end{bmatrix} \qquad B = \begin{bmatrix} 1 & 1 & 1 & 1 & 1 \\ 2 & 2 & 2 & 2 & 2 \\ 3 & 4 & 5 & 6 & 7 \end{bmatrix}.$$

If $A = U\Sigma V^*$ and $B = LDR^*$ are singular value decompositions, what are the dimensions of Σ and D? What are the last two rows of Σ? How about the last two columns of D?

2. Compute a singular value decomposition of A.

3. Equip $P_3(\mathbb{C})$ with the inner product $\langle \sum_{i=0}^{3} a_i x^i, \sum_{i=0}^{3} b_i x^i \rangle = \sum_{i=0}^{3} a_i \overline{b_j}$. Find a singular pair of bases for $T : P_3(\mathbb{C}) \to P_3(\mathbb{C})$ given by $T(p) = p'$, and find the pseudoinverse of T.

4. Suppose $\mathcal{R}, \mathcal{L}$ is a singular pair of bases for T. We showed in class that $\mathcal{R}$ is a basis of eigenvectors for T^*T. Use a similar argument to show that $\mathcal{L}$ is a basis of eigenvectors for TT^*, and compute the corresponding eigenvalues.

5. Show that T^*T and TT^* have the same eigenvalues, and that all nonzero eigenvalues have the same multiplicities. (And thus we get the same singular values for T.)

5.3 More Singular Value Decomposition

1. Consider the matrix $A = \begin{bmatrix} 1 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$.

(a) Compute a singular value decomposition of A.

(b) Verify that the columns of U and V in the singular value decomposition from part (a) form a singular pair of bases for A.

(c) Compute $A^\dagger$, the pseudoinverse of A, and compute $AA^\dagger$ and $A^\dagger A$.

2. Compute the operator norms of the following matrices.

(a) $\begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}$ **(b)** $\begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}$

3. Suppose W is a subspace of an inner product space V. Suppose $T = \text{proj}_W^\perp$ and U is an orthogonal matrix. Show that $UTU^* = \text{proj}_{U[W]}^\perp$.

4. **(a)** Find the shortest distance between the point $(1, 2, 2)$ and the plane in $\mathbb{R}^3$ given by $x + 2y + 2z = 3$.

(b) Find the shortest distance from the point $(0, 0, 0)$ to the line parameterized by $\langle 2, 0, 1\rangle + t\langle 1, 1, 0\rangle$.

5. Suppose $\mathcal{L} = \{\mathbf{v}_1, \ldots, \mathbf{v}_n\}$ is an orthonormal basis of eigenvectors of T^*T, and let λ_i be the eigenvalue of $\mathbf{v}_i$. Suppose $\lambda_i \neq 0$ for $i \leq k$, and $\lambda_i = 0$ for $j > k$.

(a) Show $\{\mathbf{v}_1, \ldots, \mathbf{v}_k\}$ is a basis for $\ker(T)^\perp = \text{im}(T^*)$.

(b) Show $\{\mathbf{v}_{k+1}, \ldots, \mathbf{v}_n\}$ is a basis for $\ker(T)$.

5.4 Jordan Canonical Form

1. Find the Jordan canonical form of the following matrices.

(a) $\begin{bmatrix} 0 & 3 & 0 \\ 0 & 0 & 3 \\ 0 & 0 & 0 \end{bmatrix}$ **(b)** $\begin{bmatrix} 0 & 0 & -1 \\ 1 & 0 & -3 \\ 0 & 1 & -3 \end{bmatrix}$

(To save you some time for (b), the JCF has a single Jordan block with eigenvalue -1.)

2. Consider the sequence given by $a_0 = 1$, $a_1 = -3$, $a_2 = 7$, and $a_n = -3a_{n-1} - 3a_{n-2} - a_{n-3}$ for all $n \geq 3$. Compute a general form for a_n in terms of n.

3. Consider a Jordan block of the form

$$J = \begin{bmatrix} \lambda & 1 & \dots & 0 \\ 0 & \lambda & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & \lambda \end{bmatrix}$$

where $\lambda \neq 0$.

(a) Let $A = \lambda I - J$. What do the first few powers of A look like?

(b) Prove that there is some n such that $A^n = 0$.

(c) Fix such an n, and compute $(\lambda I - A)(I + \lambda^{-1}A + \lambda^{-2}A^2 + \cdots + \lambda^{-n}A^n)$.

(d) Use this to find an inverse for J.

4. For an $A \in \mathbb{C}^{n \times n}$, we define $e^A = \sum_{n=0}^{\infty} \frac{1}{n!} A^n$. Show that e^J exists (i.e. the sum converges) if J is a Jordan block.

5. Compute e^A for the following values of A.

(a) $\begin{bmatrix} 0 & -4 \\ 1 & 4 \end{bmatrix}$ **(b)** $\begin{bmatrix} 0 & 1 \\ 1 & 1 \end{bmatrix}$

5.5 More Jordan Canonical Form

1. Compute the Jordan canonical form for each of the following matrices.

(a) $\begin{bmatrix} 1 & 1 & 1 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{bmatrix}$ **(b)** $\begin{bmatrix} 0 & 0 & 2 \\ 1 & 0 & -5 \\ 0 & 1 & 4 \end{bmatrix}$

2. Suppose W and U are subspaces of V such that $V = W \oplus U$. If A is a basis for W and B is a basis for U, show $A \cup B$ is a basis for V.

3. Consider a Jordan block of the form

$$J = \begin{bmatrix} \lambda & 1 & \dots & 0 \\ 0 & \lambda & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & \lambda \end{bmatrix}$$

where $\lambda \neq 0$.

(a) Let $A = \lambda I - J$. What do the first few powers of A look like?

(b) Prove that there is some n such that $A^n = 0$.

(c) Fix such an n, and compute $(\lambda I - A)(I + \lambda^{-1}A + \lambda^{-2}A^2 + \cdots + \lambda^{-n}A^n)$.

(d) Use this to find an inverse for J.

4. Compute the inverse of the matrix

$$\begin{bmatrix} 2 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 2 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 3 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 3 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 3 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 3 & 1 \\ 0 & 0 & 0 & 0 & 0 & 0 & 3 \end{bmatrix}.$$

5. For a matrix A and polynomial $p(x) = \sum_{i=0}^{n} a_i x^i$, we define $p(A) = a_0 I + \sum_{i=1}^{n} a_i A^i$.

(a) Show that, if J is a matrix in Jordan canonical form, then its characteristic polynomial χ_J satisfies $\chi_J(J) = 0$ (the zero matrix).
(Hint: Note that the standard basis is a basis of generalized eigenvectors. Write a given $\mathbf{x}$ as a linear combination of these, and analyze how the factors of the characteristic polynomial treat each term in this linear combination.)

(b) Show that for *any* square matrix A, $\chi_A(A) = 0$. This is known as the *Cayley-Hamilton* theorem.

5.6 Final Exam Review

1. Compute the minimal norm least-squares solution to the following equation.

$$\begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \\ 1 & 0 & 1 \end{bmatrix} \mathbf{x} = \begin{bmatrix} 3 \\ 4 \\ 5 \end{bmatrix}$$

2. Compute the determinant of the following matrix.

$$\begin{bmatrix} 1 & 1 & 1 & 1 \\ -2 & 2 & 2 & 2 \\ -3 & -3 & 3 & 3 \\ -4 & -4 & -4 & 4 \end{bmatrix}$$

3. Let $f : \mathbb{R}^n \to \mathbb{R}$ be linear.

(a) Show there is a unique $\mathbf{v}_f \in \mathbb{R}^n$ such that for every $\mathbf{x} \in \mathbb{R}^n$, $f(\mathbf{x}) = \langle \mathbf{v}_f, \mathbf{x} \rangle$.

(b) For such f and $\mathbf{v}_f$, how does $\ker(f)$ relate to $\mathbf{v}_f$? What is f^*?

4. **(a)** Suppose A is upper triangular and 3×3:

$$A = \begin{bmatrix} 1 & a_{12} & a_{13} \\ 0 & a_{22} & a_{23} \\ 0 & 0 & a_{33} \end{bmatrix}$$

where all entries are non-negative, $a_{ii} < 1$, and each column sums to 1. Compute $\lim_{k\to\infty} A^k$. (Hint: First show that $\lim_{k\to\infty} A^k$ has rank 1 by considering (generalized) eigenvectors of A. Then use the fact that $\mathbf{e}_1$ is an eigenvector and the fact that powers of stochastic matrices stay stochastic to compute the limit.)

(b) Can you generalize this to $n \times n$ matrices? What does this say about Markov chains?

5. Suppose that $A \in \mathbb{C}^{n\times n}$. Show that A is similar to A^T.

(Hint: Reduce the problem to showing this for Jordan blocks, and then do so.)